Nicholas C. Lopes

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EXPERIENCE

Embry-Riddle Aeronautical University Daytona Beach, FL
Graduate Research Assistant — Thermal Science Laboratory Jan 2021 – May 2026

- Performed CFD of supercritical flows & latent storage processes to evaluate relevant thermal behavior
- Mentored 10+ undergraduate and graduate researchers and established best practices for the team
- Synthesized results into 20+ publications and presented findings at technical conferences
- Wrote research proposals and secured $10,000+ in project funding

Oak Ridge National Laboratory (ORNL) Oak Ridge, TN
Thermal Storage Research Intern — Multifunctional Equipment Integration Group Jan 2025 – Dec 2025

- Led end-to-end CFD modeling of a DOE-funded heat pump water heater with embedded phase change capsules
- Developed simulation-ready CAD geometry from experimental hardware and established solver workflows
- Coordinated with ORNL researchers to translate experimental data and specifications into validated CFD models
- Performed sensitivity and parametric studies, improving phase change capsule utilization by 50+%

German Aerospace Center (DLR) Cologne, Germany
Combustion Research Intern — Institute of Propulsion Technology Jun 2024 – Aug 2024

- Designed and optimized a novel cooling hole array for a fuel-lean combustor liner
- Built a full-cycle CFD workflow using DLR's high-performance computing systems
- Collaborated with DLR experimentalists to benchmark and validate simulations against combustor test data
- Improved cooling performance by 15+% and published findings in peer-reviewed journal and conference proceedings

Pratt & Whitney West Palm Beach, FL
Aero-Thermal Engineer Intern — Aero-Thermal Fluids Group Jun 2022 – Aug 2022

- Conducted CFD/FEA of PW4000 and F-135 hot-section frames and cases
- Integrated multiple CAD models into a unified combustor-turbine assembly and defeatured geometry for analysis
- Interfaced with structures team and reported thermal profiles for life-cycle assessments
- Developed code to automate data processing and organization for auxiliary power unit systems

PROJECTS

A3 Refrigerant Leakage Analysis Jan 2026 - Jun 2026
Oak Ridge National Laboratory (ORNL)

- Evaluated A3 refrigerant leakage behavior from critical HVAC components with high-fidelity CFD
- Delivered key insights to ORNL including gas dispersion and ignition likelihood to enhance mitigation strategies

EDUCATION

Embry-Riddle Aeronautical University Daytona Beach, FL

Doctor of Philosophy in Mechanical Engineering Aug 2023 – May 2026
Master of Science in Mechanical Engineering Jan 2021 – May 2023
Bachelor of Science in Aerospace Engineering Aug 2017 – Dec 2020

SKILLS

Programming & Version Control: MATLAB, Python, C, Scheme, Perl, Git
CAD Software: SolidWorks, NX, ANSYS Spaceclaim, ANSYS Discovery, ANSYS Design Modeler
CFD & FEA Tools: ANSYS Fluent, Pointwise, ANSYS Mechanical, ANSYS Workbench, ANSYS CFX, Tecplot
High-Performance Computing: Linux environments, Torque & Slurm job schedulers, Bash/shell scripting